MD5: Applied Mathematics for Engineering (AME)

MD-5			Teaching Scheme (Hours/Week)				Credit Scheme				Examination/ Evaluation Scheme and Marks						
											Theory			Practical			Total
Sem	Course code	Name of the Course	L	P	T	Total	L	P	T	Total	CIE [20]	ISE [20]	ESE [60]	CIE TW	ESE P	ESE OR	
3	**0305151**	Linear Algebra with Python (LAP)	2	-	-	**2**	2	-	-	**2**	20	20	60	-	-	-	**100**
3	**0305251**	Linear Algebra with Python Lab (LAPL)	-	2	-	**-**	-	1	-	**1**	-	-	-	-	25		**25**
4	**0405152**	Statistical Techniques and Numerical Methods with R (STNMR)	2	-	-	**2**	2	-	-	**2**	20	20	60	-	-	-	**100**
4	**0405252**	Statistical Techniques and Numerical Methods with R Lab (STNMRL)	-	2	-	**-**	-	1	-	**1**	-	-	-	-	25		**25**
5	**0505153**	Fuzzy Logic and Graph Theory with MATLAB/ Python (FLGTM)	2	-	-	**2**	2	-	-	**2**	20	20	60	-	-	-	**100**
5	**0505253**	Fuzzy Logic and Graph Theory with MATLAB/ Python Lab (FLGTM/ PL)	-	2	-	**-**	-	1	-	**1**	-	-	-	-	25		**25**
6	**0605154**	Optimization Techniques (OT)	2	-	-	**2**	2	-	-	**2**	20	20	60	-	-	-	**100**
6	**0605254**	Optimization Techniques Lab (OTL)	-	2	-	**-**	-	1	-	**1**	-	-	-	-	25		**25**
7/8	**0805255**	Field Study/ Case Study (FS/CS)	-	4	-	**4**	-	2	-	**2**	-	-	-	50	-	-	**50**
		Total	**8**	**10**	**-**	**18**	**8**	**6**	**-**	**14**	**80**	**80**	**240**	**50**	**100**	**0**	**550**

Second Year B-Tech (SY B-Tech) AY (2025-26)

[0305151]: Linear Algebra with Python (LAP)

Semester	Credits	Teaching Scheme	Examination Scheme
3	2	Th: 2 Hrs. / Week	CIE: 20 Marks ISE: 20 Marks ESE: 60 Marks

Prerequisite: Students should have prior knowledge of

Basics of Matrices and System of Linear Equations, Statistics, Partial Differentiation

Course Objectives: The objective of this course is to provide students with

- Basic vector and matrix operations essential for ML & AI.
- How linear transformations are applied in AI
- Vector spaces and their role in ML models.
- Learn advanced techniques for feature engineering and optimization.

Course Outcomes: After completing this course, students will be able to

CO1: Explain vector and matrix operations and their role in solving linear equations. (Understand)
CO2: Differentiate between various linear transformations and their applications in ML. (Analyze)
CO3: Compute vector norms and distance metrics to measure similarity in ML models. (Apply)
CO4: Describe the significance of matrix factorization techniques in AI applications. (Understand)

COURSE CONTENTS

Module-I	Fundamentals of Linear Algebra	06 Hrs.

Vectors and Operations: Definition and representation of vectors, Vector addition, subtraction, and scalar multiplication, Dot product and its applications in ML (e.g. cosine similarity in NLP).
Matrices and Operations: Matrix addition, subtraction, scalar multiplication, Matrix multiplication and properties, Identity and inverse matrices.
Systems of Linear Equations: Solving linear equations using matrix inversion, Gaussian elimination and LU decomposition, Applications in ML: Feature scaling, linear regression.

Module-II	Vector Spaces & Norms	06 Hrs.

Vector Spaces and Subspaces: Basis and dimension, column space, row space, null space. Applications in data compression & neural networks.
Norms and Distance Metrics: Manhattan distance (L1 norm), Euclidean distance (L2 norm), and Max norm ($L\infty$ norm), Applications in k-NN, clustering, loss functions.
Orthogonality and Projections: Concept of orthogonality, Projection of vectors onto subspaces, Gram-Schmidt process.

Module-III	Transformations, Eigenvalues and Eigenvectors	06 Hrs.

Linear Transformations: Concept of transformation in ML, Rotation, scaling, and reflection. Applications in image processing and neural networks.
Eigenvalues and Eigenvectors: Definition and computation, Eigen decomposition and diagonalization, Applications in dimensionality reduction (PCA).
Singular Value Decomposition (SVD): Matrix factorization and compression. Applications in image compression, Latent Semantic Analysis (LSA) in NLP.
Covariance Matrix: Covariance matrix and correlation matrix.

Module-IV	Advanced Matrix Factorization & AI Applications	06 Hrs.

Positive Definite Matrices: Jacobian and Hessian matrix, Applications in gradient descent & neural networks.
Tensor Operations & Backpropagation: Introduction to tensors, Tensor operations in NumPy, PyTorch, and TensorFlow. Applications in CNNs, RNNs, and Transformers, Backpropagation algorithm using Jacobian and Hessian matrices.

Text Books:

T1.	"Linear Algebra and Its Applications" by Gilbert Strang, Cengage Learning, 5th Edition
T2.	"Higher Engineering Mathematics", by B. S. Grewal, Khanna Publication, New Delhi
T3.	"Python Programming" by Reema Thareja, Oxford University Press, (2nd Edition)
T4.	"Python for Data Analysis" by Wes McKinney, O'Reilly Publication, (2nd Edition)

Reference Books:

R1.	Mathematical Foundations of Machine Learning" by V. Krishna Reddy, Universities Press
R2.	"Linear Algebra and Its Applications" by T. Veerarajan, McGraw Hill Education
R3.	"Mathematics for Machine Learning" by Marc Peter Deisenroth, A. Aldo Faisal, Cheng Soon Ong, Cambridge University Press, 1st Edition, 2020
R4.	"Advanced Engineering Mathematics", by Erwin Kreyszig, Wiley India, 10th Edition.
R5.	"Higher Engineering Mathematics", by B. V. Ramana, Tata McGraw Hill.
R6.	"Machine Learning Using Python" (1st Edition, 2019) – U Dinesh Kumar Publisher: Wiley India (Book with Case studies and ML applications)
R7.	"Deep Learning for Beginners" by Rajiv Chopra, Khanna Publishing

Relevant MOOCs Course (Course name and Weblink)

3. Applied Linear Algebra in AI and ML
   https://www.youtube.com/watch?v=Ymn8rzTaEfk&list=PLyqSpQzTE6M912cneV79Cqsj88E3VBfxt
4. Applied Linear Algebra Course Introduction
   https://www.youtube.com/watch?v=Ymn8rzTaEfk&list=PLyqSpQzTE6M912cneV79Cqsj88E3VBfxt
5. Applied Linear Algebra in AI and ML by Prof. Swanand Khare | IIT Kharagpur -NPTEL
   https://www.youtube.com/watch?v=YedupuMX2to&list=PL90ZtAEZtjlmTgXn39jMCvGY3fQsMa0GJ

Relevant Topic for Self-study:

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Second Year B-Tech (SY B-Tech) AY (2025-26) **[0305251]: Linear Algebra with Python Lab (LAPL)**			
Semester	**Credits**	**Teaching Scheme**	**Examination Scheme**
3	**1**	**P: 2 Hrs./ Week**	**ESE (P): 25 Marks**

Prerequisite: Students should have prior knowledge of

- Fundamentals of programming and logic building skills.

Course Objectives: The objective of this course is to provide students with

- **Understand** Vector and Matrix Operations and Solve Linear Systems – Introduce students to fundamental vector, matrix operations and to solve linear systems using Python for mathematical computations.
- **Analyze** Distance Metrics and Orthogonality – Understand and apply distance metrics in machine learning while exploring the concept of orthogonality for data representation.
- **Apply** matrix transformations for data manipulation and analyze Singular Value Decomposition (SVD) using Python
- **Implement** Jacobian and Hessian matrices using Python for optimization problems.

Course Outcomes: After completing this course, students will be able to

CO1: Perform and analyze vector and matrix operations using Python. (Apply, Analyze)
CO2: Apply the Gram-Schmidt process to obtain an orthonormal basis. (Apply)
CO3: Use matrix transformations for effective data manipulation and analysis. (Apply)
CO4: Illustrate and Implement Jacobian and Hessian matrices using Python. (Understand, Apply)

COURSE CONTENTS

Expt. No	Title/Problem Statement	Hrs.	CO
1.	Vector Operation and Matrix Operation by using python	2	MDP5-1.1
2.	Solve linear system of equation by python	2	MDP5-1.1
3.	Distance metric for ML & Orthogonality	2	MDP5-1.2
4.	Gram - Schmidth process for Orthonormality	2	MDP5-1.2
5.	Data transformation with matrices	2	MDP5-1.3
6.	Computing singular value decomposition by using python	2	MDP5-1.3
7.	Exploring Jacobian - Hessian matrix by using python	2	MDP5-1.4
8.	Tensor Operations in AI	2	MDP5-1.4

Text Books:

T1. Linear Algebra with Python Theory and Applications
By Makoto Tsukada, Yuji Kobayashi, Hiroshi Kaneko, Sin-Ei Takahasi, Kiyoshi Shirayanagi, Masato Noguchi · 2023

T2. Linear Algebra A second course, featuring proofs and Python by Sean Fitzpatrick
University of Lethbridge

Reference Books and Links:

R1. LINEAR ALGEBRA: STEP BY STEP by Kuldeep Singh (Author)

R2.	**Numpy Library**: https://numpy.org/doc/stable/reference/routines.linalg.html
R3.	**Scipy Library** : https://docs.scipy.org/doc/scipy/reference/linalg.html
Any Special Guidelines:	

Second Year B-Tech (SY B-Tech) AY (2025-26)

[0405152]: Statistical Techniques and Numerical Methods with R (STNMR)

Semester	Credits	Teaching Scheme	Examination Scheme
4	2	Th: 2 Hrs. / Week	CIE: 20 Marks ISE: 20 Marks ESE: 60 Marks

Prerequisite: Students should have prior knowledge of

Basics Matrices and System of Linear Equations, Statistics, Partial Differentiation, Integration, Mean Value Theorems

Course Objectives: The objective of this course is to provide students with

- **Understand** and apply R programming concepts including data structures, variables, data types, and data visualization techniques to effectively manage and analyze datasets.
- **Perform** descriptive statistical analysis and hypothesis testing in R, utilizing various statistical techniques such as measures of central tendency, correlation analysis, and hypothesis testing.
- **Implement** numerical methods and solutions for systems of linear equations using R, including interpolation, root-finding techniques, and matrix computations.
- **Apply** numerical differentiation, integration, and solve ordinary differential equations (ODEs) using finite difference methods, quadrature methods, and Runge-Kutta techniques in R.

Course Outcomes: After completing this course, students will be able to

CO1: Explain R-programming for data visualization. (Understand)

CO2: Examine the statistical tests for hypothesis testing using R. (Analyze)

CO3: Describe the concepts of finite difference methods and interpolation, and **implement** numerical techniques using R for computations. (Understand, Apply)

CO4: Compute Numerical Differentiation and Integration and Solutions to Ordinary Differential Equations (ODEs) (Apply)

COURSE CONTENTS

Module-I	Introduction to R Programming and Data Visualization	06 Hrs.

Introduction to R programming: What is R?, Installing R and R Studio Overview, working in the Console, Arithmetic Operators, Logical Operations, Using Functions, Getting Help in R and Quitting R Studio, Installing and loading packages. Variables, and data types in R: Creating Variables - Numeric, Character and Logical Data - Vectors -Data Frames - Factors -Sorting Numeric, Character, and Factor Vectors - Special Values.

Data Visualization using R: Scatter Plots - Box Plots - Scatter Plots and Box- and-Whisker Plots Together - Customize plot axes, labels, add legends, and add colors.

Module-II	Statistical Methods using R	06 Hrs.

Descriptive statistics in R: Measures of central tendency - Measures of variability - Skewness and kurtosis - Summary functions, describe functions, and descriptive statistics by group

correlation, Linear Regression, Logistic Regression.
Testing of Hypothesis using R: Z-Test, t-Test, F-Test, Chi-Square test, Analysis of Variance and Correlation.

Module-III	Finite Differences, Interpolation and System of Linear Equation using R	06 Hrs.

Taylor's series, Forward, backward and central differences, Difference tables, Finite difference operators, Newton's forward and backward interpolation formulae. Lagrange's polynomial and Newton's divided difference formula.
Bisection method, Newton-Raphson method, Secant method, Implementing these methods in R.
Gaussian elimination and LU decomposition, Matrix inversion and condition numbers, using R for matrix computations.

Module-IV	Numerical Differentiation & Integration and Solutions to Ordinary Differential Equations (ODEs)	06 Hrs.

Numerical Differentiation, Trapezoidal and Simpson's rules, Adaptive quadrature methods, implementing integration techniques in R, Initial value problems, Euler's method, Modified Euler's method, Runge-Kutta methods, Solving ODEs using R.

Text Books:

T1.	Crawley, M. J. (2006), "Statistics - An introduction using R", John Wiley, London 32.
T2.	Purohit, S.G.; Gore, S.D. and Deshmukh, S.R. (2015), "Statistics using R", second edition. Narosa Publishing House, New Delhi.
T3.	Braun & Murdoch (2007), "A first course in statistical programming with R", Cambridge University Press, New Delhi.
T4.	Numerical methods for engineers by Steven C. Chapra and Raymond P. Canale. McGraw-Hill. 2015.

Reference Books:

R1.	Wasserman, L. All of statistics: a concise course in statistical inference. Vol. 26. New York: Springer, 2004.
R2.	Gareth, J., Daniela, W., Trevor, H., and Robert, T. An introduction to statistical learning: with applications in R. Spinger, 2013.
R3.	Shahababa B. (2011) , "Biostatistics with R", Springer, New York.

Relevant MOOCs Course (Course name and Weblink)

1. An Introduction to Statistical Programming Methods with R: https://smac-group.github.io/ds/
2. Introduction to R Software: https://archive.nptel.ac.in/courses/111/104/111104100/
3. Descriptive Statistics with R Software: https://archive.nptel.ac.in/courses/111/104/111104120/
4. Numerical Methods and Simulation Techniques: https://archive.nptel.ac.in/courses/115/103/115103114/
5. Numerical Methods: https://archive.nptel.ac.in/courses/111/107/111107105/

Relevant Topic for Self-study:

Second Year B-Tech (SY B-Tech) AY (2025-26)

[0405252]: Statistical Techniques and Numerical Methods with R Lab (STNMRL)

Semester	Credits	Teaching Scheme	Examination Scheme
4	1	P: 2 Hrs./ Week	ESE (P): 25 Marks

Prerequisite: Students should have prior knowledge of
Fundamentals of programming and logic building skills.

Course Objectives: The objective of this course is to provide students with

- **Understand the fundamentals of R programming**, including data structures, variables, functions, and data visualization techniques for effective data handling.
- **Apply descriptive statistics and hypothesis testing** using R to analyze datasets and interpret statistical results.
- **Implement numerical methods for interpolation and root-finding**, and solve systems of linear equations using R.
- **Utilize numerical differentiation, integration, and ODE-solving techniques** to model and solve real-world engineering problems using R.

Course Outcomes: After completing this course, students will be able to
CO1: Develop R programming skills to manipulate data structures, perform operations, and create data visualizations for analysis.
CO2: Analyze datasets using descriptive statistics and hypothesis testing techniques to extract meaningful insights.
CO3: Implement numerical methods such as interpolation, root-finding, and matrix computations to solve computational problems.
CO4: Apply numerical differentiation, integration, and ODE-solving techniques to model and analyze real-world engineering problems.

COURSE CONTENTS

Expt. No	Title/Problem Statement	Hrs.	CO
1.	Basic Operations and Data Handling in R	**2**	
2.	Data Visualization in R	2	
3.	Descriptive Statistics in R	2	
4.	Hypothesis Testing and Correlation in R	2	
5.	Interpolation Using R	2	
6.	Root Finding Algorithms and Solving System of Linear Equations in R	2	
7.	Numerical Integration in R	2	
8.	Solving Ordinary Differential Equations in R	2	

Text Books:

T1.	Crawley, M. J. (2006), "Statistics - An introduction using R", John Wiley, London 32.
T2.	Purohit, S.G.; Gore, S.D. and Deshmukh, S.R. (2015), "Statistics using R", second edition. Narosa Publishing House, New Delhi.
T3.	Braun & Murdoch (2007), "A first course in statistical programming with R",

	Cambridge University Press, New Delhi.
T4.	Numerical methods for engineers by Steven C. Chapra and Raymond P. Canale. McGraw- Hill. 2015.
Reference Books:	
R1.	Wasserman, L. All of statistics: a concise course in statistical inference. Vol. 26. New York: Springer, 2004.
R2.	Gareth, J., Daniela, W., Trevor, H., and Robert, T. An introduction to statistical learning: with applications in R. Spinger, 2013.
R3.	Shahababa B. (2011) , "Biostatistics with R", Springer, New York.
Relevant MOOCs Course (Course name and Weblink)	
1. An Introduction to Statistical Programming Methods with R: https://smac-group.github.io/ds/ 2. Introduction to R Software: https://archive.nptel.ac.in/courses/111/104/111104100/ 3. Descriptive Statistics with R Software: https://archive.nptel.ac.in/courses/111/104/111104120/ 4. Numerical Methods and Simulation Techniques: https://archive.nptel.ac.in/courses/115/103/115103114/ 5. Numerical Methods: https://archive.nptel.ac.in/courses/111/107/111107105/	